

RITESH KUMAR

Applied AI Research & Engineering | Agentic AI | NLP | Multimodal AI
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PROFILE

Applied AI researcher and technical leader with 10+ years of experience spanning **industrial research and production engineering**. Combines expertise in enterprise AI agents, document intelligence, multimodal AI, self-evolving agents and agentic RAG with a track record of published research, patented methods and contributions to technology incorporated into commercial products.

INTELLECTUAL PROPERTY

1 granted US patent + 4 filed US patent applications

RESEARCH

EMNLP 2025 | Middleware 2024 | AI4BPM 2023

GLOBAL AI CAREER

UAE | USA | Singapore

PROFESSIONAL EXPERIENCE

INCEPTION

Senior Applied Scientist

Feb 2026 - Present

Abu Dhabi, UAE

- Regulatory policy enforcement** - Design and develop a multi-agent platform that automates policy-compliance reviews for Abu Dhabi government entities, including ADAFSA and ADJD, replacing manual regulatory-review workflows with grounded, auditable AI-assisted checks. Designed and implemented self-service capabilities that support multiple regulators and use cases **without code changes**, automating policy-text extraction, vector-index creation, entity onboarding and the dynamic instantiation of specialized AI agents from declarative schemas.
- Judicial decision support system** - **Led the design and implementation** of a LangGraph-based multi-agent decision-support system for prosecutors and judges, with cooperating agents for evidence analysis, prosecution and investigation reasoning, verdict drafting and legal-citation retrieval. Established an evaluation harness across **800+ real penal cases**, assessing verdicts, penalties and UAE Federal and Abu Dhabi law citations. **Internal benchmarking reported 96% accuracy**, informing improvements to retrieval, prompts and orchestration.

IBM RESEARCH

AI Research Software Engineer

Feb 2022 - Dec 2025

New York, USA

- ProtoNER** - Developed a method that enables document-understanding systems to recognize new entity types from few labelled examples while retaining previously learned types. As **first author and co-inventor**, **led the architecture and implementation** using prototypical networks. IBM Research reports comparable new-class performance with **30 annotated examples** versus a **2,600-example** conventional fine-tuning baseline, reducing the annotation needed to extend document models. Integrated into IBM document-understanding technology; covered by **US Patent 12,554,937 B2**, granted on 17 February 2026.
- CUGA agent memory** - Co-developed a memory capability for IBM Research's open-source enterprise-agent platform that turns execution histories into **reusable procedural guidance** for subsequent tasks. Contributions covered structured trajectory processing, procedural-tip generation, preservation of task intent and evaluation-analysis tools within a collaborative 29-file change set. The CUGA platform had 867 GitHub stars and 145 forks as of 20th Sept 2026.
- Multimodal efficiency** - Co-authored **VOILA**, a framework that selects the visual fidelity needed to answer a question before retrieving and processing the image, balancing expected answer quality against acquisition cost. Evaluation across five datasets and six vision-language models reported **50-60% cost reductions** while retaining **90-95% of full-resolution accuracy**.
- Enterprise agent middleware** - Built reusable agentic middleware for a low-code/no-code enterprise agent-development framework, integrating robust implementations of ReAct, ReWOO, and Plan-and-Solve agentic architectures with review and reflection components for composable enterprise workflows. Built and delivered an AI assistant combining RAG, database search, web search, and email generation using this platform to support IBM sales teams. Components from this agentic framework subsequently became part of **IBM Watson Orchestrate**.
- FLOW-BENCH / FLOW-GEN** - Contributed to a research framework for generating business-process and decision workflows from natural-language requirements, pairing a workflow-generation method with a benchmark for evaluating LLM performance. Made substantial contributions to the constrained Python representation of workflows, construction and validation of **101 benchmark cases**, evaluation across **eight LLMs** and preparation of the public research release. Co-authored the resulting **EMNLP 2025 Industry Track** paper.

- **Multi-camera retail analytics** - Developed a real-time computer-vision platform processing multi-camera video streams to detect customers and products, maintain global trajectories across cameras, classify people/products and generate longitudinal store-analytics insights. Implemented YOLOv4-based detection, DeepSORT tracking and ResNet-18 classification with Python/OpenCV/Keras, PostgreSQL and Docker-based components supporting end-to-end experimentation and deployment. Delivered retail-analytics capabilities, including technology **deployed across multiple Verizon stores**.
- **Continuous class expansion** - Developed a content-based image-retrieval pipeline to introduce new products/classes with less retraining overhead as store inventories evolved. Internal project measurements showed a **75% reduction in retraining time**.

- **Wi-Fi-based indoor localization** - Implemented components of a passive Wi-Fi indoor-positioning system to localize mobile devices with 2-metre positioning precision indoors. Operationalized processing pipelines and backend integrations, and built behavioural analytics for dwell time, transition patterns and visit frequency.
- **Conference application - Designed and led backend development** of an AI-driven mobile application intended to improve attendee engagement at large technical conferences, building backend architecture, APIs and data services and supporting deployments including MobiSys, MobiCom, and Microsoft Faculty Summit.
- **Platform operations** - Held system and database administration responsibilities for shared LiveLabs infrastructure, including server provisioning and maintenance, fallback arrangements for key services, database backups, and user-access administration across servers and applications.

RECOGNITION AND KNOWLEDGE TRANSFER

- **IBM global TCAP recognition | 2024**. Selected as **one of 16 Significant Contributors** for January-December 2023 under the AI category of the Technical Collaboration Achievement Program, whose scope includes **IBM Research employees worldwide**. Recognized for Watson Document Understanding and Inner Source contributions.
- **IBM Employee Equity Program | 2023 and 2024**. **Selected in consecutive years** for equity awards recognizing contributions and skills. The programme recognizes employees in select roles with highly valued skills and strong performance.
- **IBM invention achievement award | May 2023**. Received the First Patent Application Invention Achievement Award for ProtoNER, linking the research contribution to IBM's formal invention recognition.
- **UAE Government Agentic AI mentorship | June 2026**. Selected as **one of 30 mentors** from Inception's applied-science team of **more than 200**. Guided use-case discovery, solution design and 90-day implementation planning in a Ministry of Cabinet Affairs-led programme involving 300+ participants from 50 federal entities.
- **Infosys Insta Award | May 2021**. Recognized for excellent attitude and commitment while designing and developing the computer-vision solution for Verizon Store Analytics use cases.
- **Independent research attention | 2026**. Trajectory-memory research was discussed by NewMind AI Journal and independently selected for a Brisbane AI Paper Reading Club session, extending attention beyond the author team and employer.

TECHNICAL EXPERTISE

Agentic AI & LLM Systems

Machine Learning / Deep Learning

Engineering & Data

LangGraph, Langfuse, RAG, Self-evolving Agents, Agentic Memory

PyTorch, Keras, scikit-learn, OpenCV, transformers/BERT, multimodal and vision-language systems

Python, PostgreSQL, Docker, AWS, Azure, REST APIs

RESEARCH AND PUBLICATIONS

Google Scholar snapshot, 2 September 2026: **94 citations, h-index 5 and i10-index 4**. Publication record spans 2016-2026; preprints are identified separately from conference and workshop publications.

- **Trajectory-Informed Memory Generation for Self-Improving Agent Systems**
2026 | arXiv preprint | arXiv:2603.10600. Co-authored research on converting execution trajectories into reusable guidance. Reported gains of up to **14.3 percentage points** on held-out AppWorld tasks and **28.5 points** on the hardest tasks; authors are listed alphabetically.
- **VOILA: Value-of-Information Guided Fidelity Selection for Cost-Aware Multimodal Question Answering**
2026 | arXiv preprint | arXiv:2602.03007. Co-developed visual-fidelity selection before retrieval; the collaborative evaluation reports 50-60% cost reductions while retaining 90-95% of full-resolution accuracy across five datasets and six vision-language models.
- **FLOW-BENCH: Towards Conversational Generation of Enterprise Workflows**
2025 | EMNLP Industry Track | Co-author. Conversational workflow generation and reproducible evaluation through a 101-case public benchmark.
- **A Conversational Assistant Framework for Automation**
2024 | ACM/IFIP Middleware Industry Track | Co-author. Research on conversational enterprise automation.
- **ProtoNER: Few Shot Incremental Learning for Named Entity Recognition using Prototypical Networks**
2023 | AI4BPM workshop | First author. Few-shot incremental learning to reduce the annotation burden of adding entity classes.
- **Alzheimer's Disease Brain MRI Classification: Challenges and Insights**
2019 | IJCAI ARIAL workshop | Co-author. Examines evaluation challenges, including subject-level separation of training and test data.
- **A Comprehensive Study of Alzheimer's Disease Classification Using Convolutional Neural Networks**
2019 | arXiv preprint | Co-author. Compares models and preprocessing under a common MRI classification evaluation framework.
- **Demo: Fusing WiFi and Video Sensing for Accurate Group Detection in Indoor Spaces**
2016 | ACM MobiSys Companion demo | Co-author. DOI: 10.1145/2938559.2938582.

PATENTS AND INVENTIONS

One granted US patent and four further US patent applications filed by IBM; named co-inventor on each.

- **Few Shot Incremental Learning for Named Entity Recognition | GRANTED**
US 12,554,937 B2; application 18/307,199; granted 17 February 2026. Inventors: Ritesh Kumar, Saurabh Goyal and Ashish Verma. Assignee: IBM.
- **Conversational Workflow Building Using Large Language Models | FILED**
US application 19/338434; filed 24 September 2025.
- **Storage and Memory Management for Multimodal AI Systems | FILED**
US application 19/321,793; filed 8 September 2025.
- **Deployment-Aware Memory Packaging for Artificial Intelligence Agents | FILED**
US application 19/742158; filed 15 July 2026.
- **Trajectory-Based Memory Generation and Contextual Retrieval | FILED**
US application 19/742178; filed 15 July 2026.

EDUCATION

M.S. Computer Science University of Massachusetts Amherst, USA, GPA 3.94/4.0	Sep 2018 - May 2020
B.Tech. Computer Science NIIT University, Rajasthan, India, GPA 8/10	Sep 2009 - Jun 2013